HATCHSIDE · TRACEABILITY MATRIX

HatchSide traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-11-05	Amish Chadha	Generated by .kit/trace.py
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HatchSide traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-004	PR-001	No entry needed	HTS-CAL-001, HTS-BLD-001	Task walkthrough for every head at a mock manhole, TRL 4; inspection of the model at TRL 3	Met on paper: every head is fitted, worked and recovered from the rim
UR-002	PR-002	Fits common openings	HTS-CAL-001, HTS-BLD-001	Setup trials on sample manholes and hatches, TRL 4; analysis (CAL-001 section 2)	Met on paper: feet on a 3.0 m circle; rim guard inner face 672 mm from the centre
UR-003	PR-003	Working depth	HTS-CAL-001, HTS-BLD-001	Winch drum capacity check and depth trial, TRL 4	Met on paper: 15 m of rope; drum capacity to confirm on purchase
UR-008	PR-004	Working load	HTS-CAL-001, HTS-BLD-001	Analysis (CAL-001 section 3), TRL 3; proof-load test at TRL 4	Met on paper: leg buckling factor 5.5 at proof with jerk (4.8 with the rope 10 degrees off vertical); rope factor 14.3 at the working load and 4.8 with the winch at stall
UR-005	PR-005	Fast setup	HTS-CAL-001, HTS-BLD-001	Timed setup trial, TRL 4; estimate (CAL-001 section 8)	Met on paper (estimate): 9.0 min
UR-006	PR-006	Portable loads	HTS-CAL-001, HTS-BLD-001, HTS-STM-001, BOM	Weigh each packed load, TRL 4; model masses (CAL-001 section 5)	Met on paper: heaviest load 23.7 kg (grab head); whole kit 153.0 kg in nine loads
UR-010	PR-007	Quick head change	HTS-CAL-001, HTS-BLD-001	Timed changeover trial, TRL 4	Met on paper (estimate): one screw-pin shackle, about 1 min
UR-004	PR-008	Grab clears silt	HTS-CAL-001, HTS-BLD-001, HTS-STM-001, HTS-OPT-001	Trial in a silt-filled test chamber, TRL 4; analysis (CAL-001 section 8)	At risk** (estimate): 10.8 L at 70 % fill but 9.7 L after the 10 % that drips back (CAL-001 section 8; STM S9); a bigger grab or a lower target is proposed to Amish (HTS-DEC-001 D16)
UR-014	PR-009	Depth sample	HTS-CAL-001, HTS-BLD-001	Trial in a layered test tank, TRL 4	Met on paper: line counter reads 0.1 m; tube centre known from the interface
UR-003, UR-004	PR-010	DomeReach to the floor	HTS-CAL-001, HTS-BLD-001	Trial at a partner digester, TRL 4; analysis (CAL-001 section 7)	Met on paper: hinge 2.6 m, tool 3.3 m below the rim
UR-009	PR-011	Corrosion resistance		Material coupon test, TRL 4	To verify at TRL 4: not verifiable on paper; galvanized steel, 6082

User requirement	Product requirement	Title	Evidence	Verification	Status
			HTS-CAL-001, HTS-BLD-001, HTS-STM-001		aluminium, A4 fasteners and rope with isolating washers
UR-007	PR-012	Value-engineering target	HTS-CAL-001	Priced BOM (`bom/bom.csv`)	Met on paper: USD 2,838, USD 2,162 under the target
UR-002	PR-013	Heads pass a small manhole	HTS-CAL-001, HTS-BLD-001	Pass each head through a 560 mm ring, TRL 4; model bounding boxes (CAL-001 section 7)	Met on paper: largest plan diagonal 525 mm (grab); DomeReach passes folded
UR-008	PR-014	Stable in use	HTS-CAL-001, HTS-BLD-001	Tilt-pull test during proof loading, TRL 4; analysis (CAL-001 section 6)	Met on paper: tips at 25.0 degrees at the safe working load
UR-012	PR-015	Guard the opening	HTS-CAL-001	Inspection of the model and build plan, TRL 3; trial at TRL 4	Met by design: BOM lines 15, 16 and 22
UR-011	PR-016	Made locally	HTS-CAL-001	Inspection of the model, BOM and build plan	Met by design: parts are named with their stock in the D3 model; BOM column make_buy
UR-013	PR-017	Winch effort	HTS-CAL-001	Analysis (CAL-001 section 4), TRL 3; crank-force measurement at TRL 4	Met on paper: about 33 W in low gear (31 N at the 250 mm crank)
UR-008	PR-018	Not for lifting people	HTS-CAL-001	Inspection of the model labels (BOM line 24)	Met by design: labels on the head and winch; retrieval head is guide-only (HTS-DDR-001, D2)

Gaps

None.